Configure IPv6 Addressing

Task 1:- Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

Aim: To assign a static IPv6 address to the primary network adapter and verify the configuration using command-line tools.

Method 1: Windows

Step 1: Open Command Prompt

- Click the Start menu.
- Type `cmd`.
- Select Run as administrator.

Step 2: Identify the Network Adapter

Execute the following command:

```
ipconfig
```

Identify your active network adapter, such as `Ethernet` or `Wi-Fi`.

Step 3: Assign a Static IPv6 Address

For an adapter named `Ethernet`, execute:

```
netsh interface ipv6 set address interface="Ethernet"
address=2001:db8:1::10
```

Note: `2001:db8:1::10` is an example address reserved for documentation. Use an IPv6 address provided by your network administrator for a real network.

Step 4: Verify the Configuration

Execute:

```
ipconfig
```

Alternatively, use:

```
netsh interface ipv6 show addresses
```

Check that the assigned IPv6 address appears under the correct network adapter.

Method 2: Linux (Kali Linux / Ubuntu)

Step 1: Open the Terminal

Open the terminal using `Ctrl + Alt + T`.

Step 2: Identify the Network Interface

Execute:

```
ip -br addr
```

Identify your active interface, such as `eth0`, `wlan0`, or `enp0s3`.

Step 3: Assign a Static IPv6 Address

For an interface named `eth0`, execute:

```
sudo ip -6 addr add 2001:db8:1::10/64 dev eth0
```

Replace `eth0` with your actual interface name.

Step 4: Verify the Configuration

Execute:

```
ip -6 addr show
```

Alternatively:

```
ifconfig
```

Check the `inet6` entry to confirm the IPv6 address.

Note: The Linux command applies a temporary address configuration that may disappear after a reboot.

Expected Output

Parameter	Example
Network Adapter	Ethernet / `eth0`
IPv6 Address	`2001:db8:1::10`
Prefix Length	`/64`
Verification Command (Windows)	`ipconfig`
Verification Command (Linux)	`ip -6 addr show`

Result

A static IPv6 address was assigned to the selected network adapter, and the configuration was verified using the appropriate command-line too

Task 2:- Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address] Use the ping6 or ping -6 command for IPv6 testing.

Aim: To configure a Unique Local IPv6 Address (ULA) beginning with `fd` and verify connectivity by pinging another device on the same LAN.

Method: Linux (Kali Linux / Ubuntu)

Step 1: Identify the network interface

Open the Terminal and execute:

```
ip -br addr
```

Identify your active network interface, such as `eth0` or `wlan0`.

Step 2: Assign a Unique Local IPv6 Address

Assign an example IPv6 address to the first computer:

```
sudo ip -6 addr add fd12:3456:789a::1/64 dev eth0
```

Replace `eth0` with your actual interface name.

Step 3: Configure the Second Device

On another Linux computer connected to the same LAN, execute:

```
sudo ip -6 addr add fd12:3456:789a::2/64 dev eth0
```

Both devices must be connected to the same LAN, and their interfaces must be active.

Step 4: Verify the IPv6 Addresses

On each computer, execute:

```
ip -6 addr show
```

Confirm that the addresses `fd12:3456:789a::1/64` and `fd12:3456:789a::2/64` appear on the correct interfaces.

Step 5: Test Connectivity

From the first computer, ping the second computer:

```
ping -6 fd12:3456:789a::2
```

Alternatively:

```
ping6 fd12:3456:789a::2
```

Press `Ctrl + C` to stop the ping test.

Step 6: Observe the Output

If the connection is successful, the output will look similar to:

```
64 bytes from fd12:3456:789a::2: icmp_seq=1 ttl=64 time=1.2 ms
64 bytes from fd12:3456:789a::2: icmp_seq=2 ttl=64 time=1.0 ms
```

The exact response time may differ.

Expected Results

Parameter	Computer 1	Computer 2
IPv6 Address	`fd12:3456:789a::1/64`	`fd12:3456:789a::2/64`
Address Type	Unique Local IPv6	Unique Local IPv6
Test Command	`ping -6 fd12:3456:789a::2`	`ping -6 fd12:3456:789a::1`

Task 3:- Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.Document the steps and the observed address behavior in a text file.

Aim: To configure a temporary IPv6 address and check whether it changes after restarting the network or computer.

Step 1: Open the Linux terminal and check the network interface.

```
ip -br addr
```

Step 2: Assign a temporary IPv6 address. Replace `eth0` with your interface name.

```
sudo ip -6 addr add fd12:3456:789a::10/64 dev eth0
```

Step 3: Verify the IPv6 address.

```
ip -6 addr show
```

Step 4: Disconnect and reconnect the network, then check again.

```
ip -6 addr show
```

Step 5: Restart the computer and check the address again.

```
ip -6 addr show
```

Task 4:- Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

```
restaurants = [
    {"name": "Pizza Palace", "ipv6": "2001:db8::1"},
    {"name": "Burger House", "ipv6": "2001:db8::2"},
    {"name": "Spice Garden", "ipv6": "2001:db8::3"},
    {"name": "Food Junction", "ipv6": "2001:db8::4"}
]

search = input("Enter restaurant name: ").lower()

found = False

for restaurant in restaurants:
    if search in restaurant["name"].lower():
        print("Restaurant:", restaurant["name"])
        print("IPv6 Address:", restaurant["ipv6"])
        found = True

if not found:
    print("Restaurant not found!")
```

Task 5:- Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

Aim: To create an IPv6 subnetting scenario using AI and assign an IPv6 address from one subnet to a system.

1. AI-Generated IPv6 Subnetting Scenario

A college network uses the IPv6 prefix `2001:db8:abcd::/48`. The network administrator divides it into three subnets for different departments.

Department	IPv6 Subnet	Example Host Address
Computer Department	`2001:db8:abcd:1::/64`	`2001:db8:abcd:1::10/64`
IT Department	`2001:db8:abcd:2::/64`	`2001:db8:abcd:2::10/64`
Administration	`2001:db8:abcd:3::/64`	`2001:db8:abcd:3::10/64`

These are documentation-only IPv6 prefixes, suitable for a lab exercise rather than real internet connectivity.

2. Configuration Steps (Linux/Kali Linux)

Step 1: Check the network interface

```
ip -br addr
```

Step 2: Assign an IPv6 address

For example, assign an address from the Computer Department subnet:

```
sudo ip -6 addr add 2001:db8:abcd:1::10/64 dev eth0
```

Replace `eth0` with your actual interface name.

Step 3: Verify the address

```
ip -6 addr show
```

Check that `2001:db8:abcd:1::10/64` appears in the output.

3. Observation

The IPv6 address was assigned to the selected network interface. The configuration was checked using the `ip -6 addr show` command.

4. Result

An AI-generated IPv6 subnetting scenario containing three subnets was created, and an IPv6 address was assigned and verified on the system.

Important: The address is temporary and normally disappears after a reboot. Since the prefix is reserved for documentation, use an actual IPv6 prefix assigned to your LAN if real network connectivity is required.